Jiaqi Kevin Yang

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Education

McGill University Expected Graduation: Dec 2027
Master of Science in Computer Science *Montreal, QC*

McGill University Sep 2022 - May 2026
Bachelor of Science in Software Engineering, GPA: 3.74/4.00 *Montreal, QC*

Experience

Teaching Assistant Aug 2026 - Present
McGill University *Montreal, QC*

- Mentored students in circuits, MIPS assembly and computer architecture
- Hosted office hours to help **200+** students with technical questions, providing targeted low-level code reviews

QA Automation Intern May 2025 - Aug 2025
dormakaba *Montreal, QC*

- Built a custom Locust load testing framework that simulated up to **1000** concurrent users and tested how the backend API behaved under different stress patterns to uncover bottlenecks and scalability issues
- Eliminated over **20** hours of manual regression and integration testing per release cycle by extending a RESTful Flask API on a Raspberry Pi-powered hardware testing jig
- Removed hardcoded credentials across **3** repositories by integrating AWS SSO and Parameter Store
- Automated over **50** Java end-to-end validation tests into CI/CD pipelines across firmware and cloud releases

Projects

Distributed Middleware System | *Java, TCP Sockets, Multi-threading* Sep 2025 - Oct 2025

- Engineered a distributed system in Java with a middleware server to route concurrent client requests to replicated multi-threaded backend resource managers through a simulated **10** machine distributed network
- Developed a non-blocking TCP socket protocol, enabling async messages to maximize request throughput and prevent thread starvation under concurrent loads
- Designed atomic remote procedure calls to guarantee shared-memory states with no data races

Smart Dumbbell | *Python, Flask, MQTT, SQLite3, Multi-threading* Nov 2025

- Engineered a non-blocking, multi-threaded Flask backend to process concurrent MQTT hardware streams and a computer vision pipeline to accurately detect and classify different workout exercises
- Designed SQLite3 schemas and REST endpoints to ingest high-frequency sensor telemetry, streaming real-time rep classifications to a live HTML/CSS/JavaScript web dashboard

C Compiler | *Java, MIPS Assembly, C, Compiler Design* Jan 2026 - May 2026

- Built an end-to-end C to MIPS assembly compiler in Java, refactoring C grammar to eliminate left recursion and developing a recursive-descent parser to generate abstract syntax trees
- Engineered multi-pass semantic analysis engine to enforce strong typing, nested scoping and struct alignment alongside a code generator managing calling conventions, stack frame layout and jump offsets
- Implemented Chaitin's graph coloring algorithm to allocate registers with heuristic register spilling to memory

Ataxx Game Playing AI | *Python, MiniMax* Nov 2025 - Dec 2025

- Constructed an adversarial game-playing AI for Ataxx using minimax search with alpha-beta pruning and iterative deepening under a strict two second decision limit
- Implemented root-level heuristic move ordering, cutting depth-4 search latency by nearly **50%** to consistently reach search depths up to 7 in the decision tree
- Reduced search-tree branching by **33%** by applying move folding logic that pruned redundant state transitions

Technical Skills

Programming Languages: Java, Python, C/C++, C#, SQL, JavaScript/TypeScript, HTML/CSS, Bash
Technical Skills: Git, REST APIs, AWS, Node.js, Flask, React, pytest, Linux, Agile, Docker, Postman, Claude Code
CS Fundamentals: Concurrent Programming, Machine Learning, Test-Driven Development Operating Systems
Languages: Fluent English, Fluent French, Advanced Chinese, Beginner Japanese